Carl Emmanuel M. Macabales

0956 389 3104 | cemmacabales@mymail.mapua.edu.ph | cemmacabales.tech | linkedin/cemmacabales

Education

Mapùa University — Makati, Philippines
Bachelor of Science in Computer Science, Specializing in Artificial Intelligence — *Aug. 2023 – Present*

Publications

Multi-Class Kidney Abnormality Segmentation — Aug. 2025 – Jan. 2026
AI - Deep Learning Computer Vision | ICIPCN 2026

- Developed a YOLOv12-based detection model for identifying kidney cysts, stones, and tumors from CT scans.
- Curated and processed a dataset of **14,761 CT images** with augmentation to improve model generalization.
- Achieved **mAP@0.5 of 0.946** (axial) and **0.885** (coronal) across kidney abnormality classes.
- Demonstrated potential to reduce radiologist workload and improve diagnostic consistency via automation.
- Presented at the 6th International Conference on Image Processing and Capsule Networks, Kathmandu University, Nepal.

RAG-Based Clinical Guideline Chatbot for Atrial Fibrillation — Dec. 2025 – Mar. 2026
AI – Natural Language Processing | CSPA 2026

- Built a RAG chatbot with Llama-3, Phi-3, and Qwen3 for querying **100+ pages** of ESC AF guidelines.
- Improved retrieval via MedCPT, FAISS, and BGE reranking; achieved **BERTScore F1 0.835** and **ROUGE-1 0.456**.
- Evaluated on **20 clinical queries**, reaching faithfulness up to **8.75/10** with hallucination case analysis.
- Deployed quantized LLMs achieving ∼**0.7s latency** and up to **7.9 tokens/sec** throughput.

Projects

Real-Time AI Exercise Coaching System — Apr. – Jun. 2026
AI – Computer Vision | Python, MediaPipe BlazePose, TFLite, ONNX, Raspberry Pi 5

- Trained a dual-head LSTM on **451,638** augmented **30-frame windows** from **174 videos**, achieving **97.15%** exercise classification and **92.81%** form quality accuracy across **9 movement patterns**.
- Exported to TFLite (**219 KB**) with BlazePose Lite for edge deployment, enabling **25–30 FPS** real-time inference on a Raspberry Pi 5.
- Built an on-device RAG coaching chatbot (∼**115 MB** footprint) using an ONNX sentence-transformer over pre-compiled PDF fitness manuals, eliminating PyTorch from the Pi runtime entirely.

iPrayUST – Digital Prayer Companion — Jun. 2025 – Dec. 2025
Mobile App Development | React Native, Expo, TypeScript, Firebase

- Developed a cross-platform mobile app for the UST community supporting **10,000+ users** with daily content.
- Built a Firebase backend (Firestore, Auth, Storage) managing **500+ prayer resources** and an admin CMS reducing content update time by ∼**80%**.
- Implemented offline-first caching, cutting network requests by ∼**40%** and maintaining **60fps** UI performance.

Activities

Pink Raft — 1st Runner-Up, Stellar Hackathon 2026 — May 2026
Full-Stack Engineer & AI Integration Lead

- Engineered a no-code visual payment flow builder on Stellar/Soroban, letting non-technical users drag-and-drop triggers/actions and deploy live smart contracts in under **60 seconds**.
- Cut end-to-end AI response latency from ∼**10s** to ∼**2s** (**80% reduction**) via null-stripping, Zod schema hardening, and trigger-conflict detection replacing silent failures with structured responses.
- Shipped production auth, WebAuthn 2FA, non-custodial wallet signing (Freighter/xBull/Albedo), and live on-chain event feed using Next.js 15, PostgreSQL, Redis, and Soroban smart contracts on Railway CI/CD.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML/CSS, R
Frameworks & Libraries: React, React Native, Flask, FastAPI, Expo, Node.js, SCSS
AI/ML: RAG, NLP, Computer Vision, Object Detection, PyTorch, OpenCV, NumPy, pandas
Developer Tools: Git, Docker, Firebase, Firestore, Streamlit, VS Code, NVIDIA CUDA
Cloud & Databases: Firebase, Appwrite, PostgreSQL, Google Cloud Platform, RESTful APIs